

# Baselines First

Project: Responsible Plant Classifier

## 1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Dataset Cards

Without opening the old lesson, explain one key idea from Dataset Cards. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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### Sampling and Intended Use

Without opening the old lesson, explain one key idea from Sampling and Intended Use. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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### Python and Table Readiness

Without opening the old lesson, explain one key idea from Python and Table Readiness. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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## 2. Practise with fading help

Use W2: Calculate the validation score for A baseline before a model. Record the answer once; the lesson and workbook refer to the same attempt.

**Given inputs · Procedure I chose · Middle steps · Result and check**

**Given inputs:** \_\_\_\_\_

**Procedure I chose:** \_\_\_\_\_

**Middle steps:** \_\_\_\_\_

**Result and check:** \_\_\_\_\_

\_\_\_\_\_

**The step I completed without help, and the hint I needed**

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## 3. Individual reasoning check

Use the worked case for A baseline before a model. Proposed explanation: “A simple baseline does not need training data”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

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## 4. Independent transfer case

## Independent case: Fit and compare independently

New training labels are blocked, clear, blocked, blocked, clear. New validation labels are clear, blocked, clear, blocked. Fit the same baseline rule and list its four predictions. A supplied candidate predicts clear, blocked, blocked, blocked. Score both on these rows and state the difference in correct examples and percentage points. What claim remains unsupported?

### My independent explanation, trace and limits

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

## 5. Your learning target

A dataset contains 27 healthy and 3 spotted leaves. An always-healthy baseline gets what accuracy and spotted recall? What must a new model improve upon?

## 6. Solve the given case

A test set has eighteen healthy and two spotted leaves. Baseline always predicts healthy.

Calculate and critique accuracy.

Show your trace, calculation or evidence

### Write or draw your reasoning and evidence.

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## 7. Check a changed case

A dataset has 18 grass and 2 fern examples. A baseline always predicts grass. Compute accuracy and name a serious limitation.

My independent answer and reason

**Write or draw your reasoning and evidence.**

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### Week 10 Test and reflect

## 8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

### Test record · My evidence

**Test record:** Changed input or condition

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Test record:** Expected answer or fallback

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Test record:** Observed result

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Test record:** Why the result occurred

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

## 9. Project checkpoint

Record majority and transparent-rule baselines for the Plant Classifier using the frozen test plan.

The project change and evidence I will keep

**Write or draw your reasoning and evidence.**

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## 10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review \_\_\_\_\_ Next step \_\_\_\_\_

## 11. Transfer practice from the original programme

A dataset contains 27 healthy and 3 spotted leaves. An always-healthy baseline gets what accuracy and spotted recall? What must a new model improve upon?

**My independently reasoned answer**

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Record support used: none, hint or teacher explanation.

## 12. W1 Choose from training evidence

Count clear and blocked in the worked training list. State the constant prediction and the declared tie rule. Explain why the validation majority does not choose this baseline.

**My reasoning and evidence**

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**13. W2 Calculate the validation score**

List the four baseline predictions and mark which validation rows are correct. Give correct/total and accuracy. Explain why 4/6 is not the answer.

**My reasoning and evidence**

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**14. W3 Make the comparison fair**

A candidate is scored on ten easy rows and the baseline on four difficult rows. Describe a fair comparison and two details you must record.

**My reasoning and evidence**

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## 15. W4 Fit and compare independently

New training labels are blocked, clear, blocked, blocked, clear. New validation labels are clear, blocked, clear, blocked. Fit the same baseline rule and list its four predictions. A supplied candidate predicts clear, blocked, blocked, blocked. Score both on these rows and state the difference in correct examples and percentage points. What claim remains unsupported?

### My reasoning and evidence

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## 16. W5 Explain the hidden error

A training-derived clear baseline gets nine of ten validation cases correct; the labels are nine clear and one blocked. State its accuracy and missed case. If a candidate also gets nine correct, must it miss the same case? Explain.

### My reasoning and evidence

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## 17. Feedback, return check and learning record

### Evidence · My record

**Evidence:** Worked alone / hint / model

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** First step I repaired and why

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** New case checked later; date and result

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** What I can now explain without notes

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** One remaining question

**My record:** \_\_\_\_\_

\_\_\_\_\_

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

## 18. Pictorial case practice: Baselines First

Make a simple baseline earn its place before claiming a better classifier.

Training labels: healthy, healthy, spotted. Majority baseline predicts healthy. Validation references: healthy, spotted, healthy, spotted.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

## Baselines First

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



### 1. Fit baseline

Training majority is healthy: 2 of 3 labels.



### 2. Score baseline

Always healthy matches 2/4 validation cases: 50%.



### 3. Compare candidate

Supplied candidate predictions healthy, spotted, spotted, spotted match 3/4: 75%.

### CHANGE ONE THING

Change the third candidate prediction from spotted to healthy.

Explain the changed result using the matching lesson.

Can the candidate's training accuracy replace this validation comparison? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

### My picture-based explanation and changed-case prediction

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Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

## 19. Project portfolio milestone

Project: Responsible Plant Classifier

Current milestone: Make a transparent first version. Keep a trace of how one input becomes an output; explain the rule or learned component.

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

### Evidence to keep · My record

**Evidence to keep:** This week's input and deciding step

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence to keep:** Prediction and actual result (or labelled paper trace)

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence to keep:** One change since the previous checkpoint and why

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence to keep:** One error or limit and the next test

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Explain what this week adds to your project. Reuse last week's record rather than starting again.**

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